Two Varieties, Two Sarcasms: A Cross-Validated Ensemble and an Analysis of Australian and British Sarcasm Labels

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Abstract

We describe the system of team Stargazers (CodaBench user `thermostatic`) for the shared task of the Australasian Language Technology Association (ALTA) 2026 workshop: binary sentiment and sarcasm classification of Australian (en-AU) and British (en-UK) English Google reviews and Reddit comments, scored by averaging, over the two tasks, the lower of the two varieties' macro-F1. Using five-fold cross-validation, we selected a fine-tuned DeBERTa-v3-large classifier and two logistic-regression probes on frozen model representations, and we average their predicted log-odds. It scored 0.867 on the test set, ranked second, and was one of six joint winners. Doubling the training data, or averaging more DeBERTa training runs, raises the cross-validated scores of our two strongest models by at most 0.004. The two varieties' sarcasm labels, however, differ. Among Reddit comments that a sentiment classifier labels negative, each en-AU subreddit has a higher share labelled sarcastic (44–64%) than each en-UK subreddit (12–33%). Classifier scores for derision (mocking, contempt, anger, cynicism) are positively associated with the en-AU sarcasm label (mean AUC 0.70) and inversely with the en-UK label (0.43). With one annotator per variety, we cannot attribute these differences to the annotators, the communities or the texts.

1 Introduction

The ALTA 2026 shared task (Mollá et al., 2026) uses the Australian (en-AU) and British (en-UK) English portions of BESSTIE, a benchmark for sentiment and sarcasm classification across varieties of English (Srirag et al., 2025). Each text is a Google review or a Reddit comment with two binary labels: sentiment (positive or negative) and sarcasm (sarcastic or not). Each task is scored by macro-F1, the unweighted mean of the two classes' F1 scores (the harmonic mean of precision and recall),

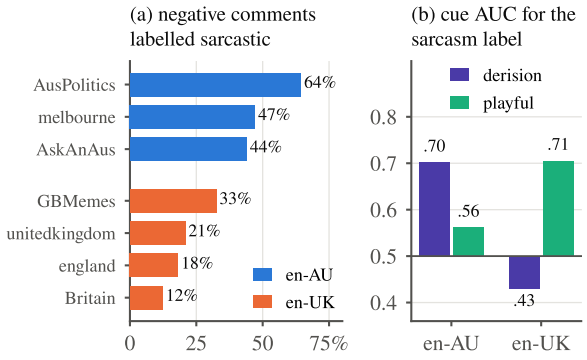


Figure 1: Sarcasm-label rates and cue associations differ between the en-AU and en-UK data. (a) Share of Reddit comments labelled sarcastic by the dataset annotator, per subreddit, among comments that the SiEBERT sentiment classifier labels negative and that we matched to the Arctic Shift archive (1,221 en-AU, 883 en-UK). AusPolitics = AustralianPolitics, AskAnAus = AskAnAustralian, GBMemes = GreatBritishMemes. (b) Mean ROC AUC of the four derision cues and the four playful cues (§6.2) for the annotator's sarcasm label, among all Reddit comments in the pool that SiEBERT labels negative, including those not matched to the archive (1,271 en-AU, 912 en-UK). ROC AUC (area under the receiver operating characteristic curve) 0.5 is chance level; values below 0.5 indicate an inverse association. No panel uses the LLM re-annotation of Appendix B.

computed separately for each variety. The official score takes the lower variety-level macro-F1 for each task and averages these two values. We call the variety with the lower macro-F1 the *binding* variety, because only its score enters the official score. For example, our test macro-F1 scores of 0.932 (en-AU) and 0.976 (en-UK) for sentiment and 0.801 (en-AU) and 0.837 (en-UK) for sarcasm give $(0.932 + 0.801)/2 \approx 0.867$.

This paper makes two contributions. First, it describes a three-member ensemble, selected with the protocol of §3, that ranked second by official score and was declared one of six joint winners. Second, it shows that the en-AU and en-UK sarcasm

labels differ in their rates and in their associations with automatically scored cues. This matters for how the benchmark is used: when a model trained on one variety scores lower on the other, part of the drop may reflect different labelling conventions or different texts rather than the model's handling of the variety. Figure 1 summarises the evidence: among comments labelled negative by a sentiment classifier, 44–64% are labelled sarcastic in each en-AU subreddit against 12–33% in each en-UK subreddit, and contempt and anger scores are associated with the sarcastic label in en-AU but with the non-sarcastic label in en-UK. These analyses describe where the labels differ; they do not identify why the official score changes little across the configurations we tested.

Our findings are:

1. The ensemble scored 0.867 on the test set, ranked second, and was one of six joint winners (§5).
2. In cross-validation, from 50% to 100% of each training fold, DeBERTa's official score rises from 0.858 to 0.862 and the large language model (LLM) probe's stays at 0.853; from three to ten averaged DeBERTa training runs, the score rises from 0.863 to 0.864. Adding the two probes to the averaged DeBERTa runs raises it by 0.006–0.007 (§6.1).
3. Derision cues are associated with the sarcastic label in en-AU (AUC 0.66–0.73), whereas contempt and anger are associated with the non-sarcastic label in en-UK (AUCs 0.41 and 0.38 across all en-UK comments that a sentiment classifier labels negative; 0.33–0.47 within each en-UK subreddit) (§6.2).
4. BESSTIE reports human agreement (Cohen's κ) of 0.47 for en-AU sarcasm; among Reddit comments with a negative gold sentiment label, our ensemble's out-of-fold sarcasm AUC is 0.77 in en-AU against 0.91 in en-UK (§6.3).

We also report interventions that did not meet our improvement criteria (§7) and an exploratory re-annotation by an LLM without access to the gold labels (Appendix B).

Related work. Two findings on sarcasm interpretation bear on ours: Oprea and Magdy (2020a) show that the sarcasm an author intends differs from the sarcasm readers perceive, and Oprea and Magdy (2020b) relate sarcasm interpretation to readers' country, age and whether English is their native language. BESSTIE extends variety benchmarks such as DIALECTBENCH (Faisal et al., 2024) to sentiment and sarcasm in English varieties, and its authors reported lower sarcasm F-scores when models are trained on one variety and evaluated on another (Srirag et al., 2025). Singh et al. (2025) add explanations to BESSTIE's Australian and Indian English sarcasm data and evaluate a prompting method (pragmatic metacognitive prompting) for explainable sarcasm detection. Our analysis concerns label variation and annotator effects (Geva et al., 2019; Basile et al., 2021; Plank, 2022). Röttger et al. (2022) distinguish descriptive annotation, which elicits annotators' own judgements, from prescriptive annotation, which asks them to apply one specified interpretation. The EPIC study collected irony judgements from annotators in five English-speaking countries, including the UK and Australia, and modelled the differences between these groups (Frenda et al., 2023). In pragmatics, jocular mockery and abuse (remarks that appear hostile but are used playfully, often to signal closeness) have been described in both Australian and British interaction (Haugh and Bousfield, 2012). Our contribution is specific to this benchmark: a system for the shared task, and an analysis of how the two BESSTIE sarcasm label sets relate to automatically scored cues, conducted within each subreddit.

2 Task and Data

The training and development files together (the *pool*) contain 6,288 texts: 3,080 en-AU (1,076 reviews, 2,004 Reddit comments) and 3,208 en-UK (2,065 reviews, 1,143 comments). We call each variety–source combination a *segment*. Of the 1,163 texts labelled sarcastic, 1,089 (94%) are Reddit comments: 42% of en-AU and 22% of en-UK comments are labelled sarcastic, against 6.8% of en-AU reviews and 1 of the 2,065 en-UK reviews. The test set contains 1,573 texts with the same segment proportions as the pool (270 en-AU reviews, 501 en-AU comments, 516 en-UK reviews, 286 en-UK comments).

BESSTIE selected Reddit comments from variety-specific subreddits and Google reviews of places in each country, so the variety tags are dataset assignments, not verified speaker identities; annotators discarded texts they judged to express no clear positive or negative sentiment. One annotator labelled each variety, and a second annotator labelled 50 items per variety for an agreement

	Comment (paraphrased)	Gold	Ours
Derision labelled sarcastic (en-AU)			
AU	A private-school boy playing gangster. Couldn't be more embarrassing if he tried.	S	S
AU	Couldn't have happened to a more deserving creep. Good riddance.	S	S
Irony labelled sarcastic (en-UK)			
UK	A sensible policy from *that* party? Has Hell frozen over?	S	S
UK	I, for one, welcome our new pigeon overlords.	S	S
Hostility labelled not sarcastic (en-UK)			
UK	How did this clown end up running the country? Total knuckle-dragger.	–	–
UK	Can we stop giving this con artist airtime? Half the country already fell for him once.	–	–

Table 1: Author-selected examples of the two readings of the guideline (not a random sample). Each row paraphrases a dataset comment about a public figure. Gold: the dataset annotator's label; Ours: our ensemble's label; S = sarcastic, – = not sarcastic.

study. The guideline reads: "Annotate if the comment uses irony or mockery to express contempt or ridicule," and asks annotators to "consider the tone and context of the comment" (Srirag et al., 2025). We see two possible readings: an inclusive one, in which literal mockery that expresses contempt or ridicule counts as sarcasm, and a narrower one, which requires non-literal (ironic) meaning. Table 1 shows examples of both readings.

The official split assigns individual Reddit comments, not whole threads, to the training and development files. We matched 95% of the Reddit texts (95.0% en-AU, 95.9% en-UK) to the Arctic Shift archive of public Reddit comments (Heitmann, 2024). The matched comments come from seven subreddits and were posted between 18 February and 2 June 2024, and 82% of the matched development comments share a thread with a training comment. With cross-validation folds that keep each thread in one fold, the ensemble's score (with DeBERTa averaged over three seeds) falls by 0.0012, from 0.8706 to 0.8694; the embedding probe falls by 0.008, DeBERTa by 0.001, and the LLM probe rises by 0.007. In this audit, the comment-level split changes the ensemble's estimate by 0.0012.

3 Validation Protocol

With 6,288 texts, labels from one annotator per variety, and a metric that takes the lower of two varieties, score differences between configurations can reflect sampling variation, training randomness or repeated selection rather than real improvements. We specified the following procedures before selecting the final ensemble; each addresses one source of error.

- **Overfitting the development file.** We pool the training and development data and split the pool into five folds, stratified jointly by variety, source, sentiment label and sarcasm label, with identical texts kept in one fold. Members and ensembles are compared on out-of-fold (OOF) predictions for all 6,288 texts, where each text is predicted by models trained without its fold. Scores on the official development file are reported only as a diagnostic.
- **Optimistic thresholds.** We *cross-fit* decision thresholds: one threshold per task and segment is fitted on four folds and applied to the fifth, so reported scores do not reuse the predictions used to choose the thresholds. For the test set, thresholds are fitted on all OOF predictions.
- **Mistaking noise for gains.** We compare two candidates with a paired bootstrap (Efron, 1979; Koehn, 2004): 2,000 resamples of the pool's OOF predictions, each the size of the official development file (757 texts; one fold's size when screening on a single fold) and stratified jointly by variety, source and both labels, with both candidates scored on the same resample and with predictions and thresholds held fixed. P_{boot} is the share of resamples in which the first candidate obtains the higher official score. It reflects sampling of the evaluation set only, not retraining or selection, and it is not a p-value: in a calibration run with a multilayer perceptron (MLP) classifier on frozen BGE-large embeddings (Xiao et al., 2024), 18% of comparisons between training runs that differ only in their random seed reached a bootstrap superiority of at least 0.9 in one direction. We treat $P_{\text{boot}} < 0.8$ as no evidence and $0.8 \leq P_{\text{boot}} < 0.95$ as suggestive evidence, and we adopt an intervention only if it beats its baseline with $P_{\text{boot}} \geq 0.8$. Because fine-tuning results vary with the random seed (Dodge et al., 2020), we compare fine-tuned encoders on predictions averaged over at least two seeds; exceptions are noted where they occur.
- **Leakage in our pipeline.** Each final member is re-trained with shuffled labels and must score within 0.02 of uniformly random probabilities with the same thresholds (official score 0.641 on the pool); DeBERTa, the embedding probe and

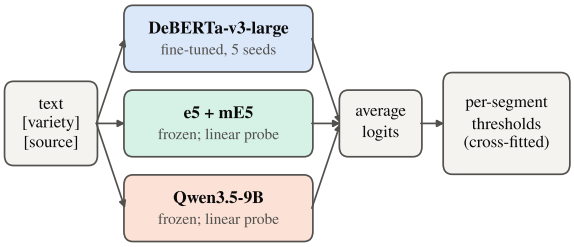


Figure 2: The submitted system. Each of three members outputs a probability per task; we convert the probabilities to log-odds, average them with equal weights, and apply one threshold per task and segment, fitted on out-of-fold predictions. En-UK reviews are always labelled not sarcastic.

the LLM probe score 0.631, 0.638 and 0.642.

- **Selection on one split.** System rankings can depend on the split (Gorman and Bedrick, 2019). Neural training configurations are therefore screened on one fold, and a configuration that improves there must also improve on a fold not used for screening, against a success criterion written down before the confirmation run (cf. Nosek et al., 2018); these criteria are internal documents, not public pre-registrations.

4 System

The system (Figure 2) combines a fine-tuned text encoder with two classifiers trained on frozen representations.

Fine-tuned encoder. We fine-tune DeBERTa-v3-large (He et al., 2023) for four epochs (encoder learning rate 10^{-5}, batch size 16). Each input is prefixed with its variety and source, e.g. `[en-UK] [reddit]`. The model has one sentiment output and two sarcasm outputs, one for each variety, trained with a learning rate of 10^{-4}. For each of the five folds we train five models with different random seeds, and we average the predicted probabilities of the 25 models on the test set.

Embedding probe. Logistic regression (Pedregosa et al., 2011) on concatenated sentence embeddings from e5-large-v2 (Wang et al., 2022) and multilingual-e5-large-instruct (Wang et al., 2024), which are not fine-tuned: one sentiment model for all segments, with segment indicators, and one sarcasm model per segment. The regularisation strength is chosen by three-fold cross-validation within each training fold.

	Sentiment		Sarcasm		
	AU	UK	AU	UK	Score
DeBERTa (5 seeds), OOF	.931	.972	.803	.816	.867
Embedding probe, OOF	.917	.969	.777	.788	.847
9B LLM probe, OOF	.916	.973	.789	.806	.853
Ensemble, OOF	.934	.976	.805	.841	.869
Ensemble, **test**	.932	.976	.801	.837	**.867**
DeBERTa (5 seeds), **test**	.916	.975	.793	.820	.854
Organisers' baseline, test	.866	.931	.680	.696	.773

Table 2: Macro-F1 for each task and variety; official score $= \frac{1}{2}[\min(\text{sentiment AU, UK}) + \min(\text{sarcasm AU, UK})]$. OOF: out-of-fold predictions on the pool, with cross-fitted thresholds (§3). The baseline is the organisers' fine-tuned DistilBERT (Sanh et al., 2019; Srirag, 2026); its test scores are from the official leaderboard.

LLM probe. Qwen3.5-9B (Qwen Team, 2026), a 9-billion-parameter LLM with weights quantised to 4 bits (Dettmers et al., 2023), processes each text inside a prompt that asks whether the text is sarcastic (or positive) and contains no labelled examples. The sarcasm prompt's definition differs by variety (Appendix C). We train a logistic regression, i.e. a linear probe (Alain and Bengio, 2017), on the layer-24 hidden-state vector at the last prompt token, concatenated with the answer margin (the pooled logits of "Yes" minus "No" for sarcasm, of "Positive" minus "Negative" for sentiment) and indicators for en-UK, Reddit and their interaction. Each variety has its own sarcasm probe.

Combination. We convert each member's probability to log-odds and average the three values with equal weights. Learned weights and stacking (Wolpert, 1992), i.e. logistic regressions trained on the members' OOF predictions, did not meet our adoption criterion. Because only 1 of the 2,065 en-UK reviews in the pool is labelled sarcastic, the system labels every en-UK review not sarcastic.

5 Results

Test score and pre-submission estimate. The ensemble scored 0.8668 on the test set (Table 2), 0.0025 below its OOF estimate of 0.8693. Before submitting, we computed a heuristic interval of 0.844–0.895: the OOF score ± 1.645 bootstrap standard deviations, widened by 20% as recommended by an internal review of the validation code; its coverage is not established. Each of the ensemble's four test macro-F1 scores differs from its OOF value by at most 0.004. En-AU had the

lower macro-F1 in both tasks, so en-AU set the official score. The test–OOF difference of 0.0025 is smaller than the bootstrap standard deviation of 0.013 (at development-file size).

Ranking. Our submission ranked second, 0.0007 below the top submission. The organisers compared the top submissions with an approximate randomisation test with 10,000 replicates (Dror et al., 2018), found no significant differences among the top six at their chosen significance level, and declared all six joint winners (Mollá et al., 2026). A non-significant difference does not show that two submissions are equivalent, only that the test did not distinguish them. According to the organisers, our submission and the top submission differ on 33 of the 1,573 sentiment labels (2.1%) and 74 of the 1,573 sarcasm labels (4.7%).

Effect of the probes on the test set. To measure the contribution of the probes, we also submitted the DeBERTa models alone. They scored 0.854, against 0.867 for the ensemble, a difference of 0.0127. The paired bootstrap on the pool had estimated a mean gain of +0.005 (90% interval −0.010 to +0.023), so the observed gain lies within that interval. On the pool, the submitted five DeBERTa seeds scored 0.867 and the ensemble 0.869; averaged over subsets of five of our ten DeBERTa training runs, adding the probes raises the score by 0.0067 (Figure 3b). On the test set, DeBERTa alone lost most in the two binding en-AU components: sentiment fell from 0.931 (OOF) to 0.916 and sarcasm from 0.803 to 0.793.

6 Training Data, Seeds and Labels

On the screening fold, the alternative encoders we tried scored 0.009–0.016 below DeBERTa-v3-large, and none of the additional interventions met our improvement criteria (§7). This section examines three factors in turn: training-set size, seed averaging and the labels.

6.1 Training-set size and seed averaging

Training-set size. With the same two seeds (13 and 14) at every size, DeBERTa's mean score over the two seeds is 0.858, 0.859 and 0.862 at 50, 75 and 100% of each training fold, a gain of 0.004; the LLM probe scores 0.853, 0.854 and 0.853 (Figure 3a). DeBERTa is trained for four epochs at every size, so smaller subsets also receive fewer parameter updates. From 50 to 100%, DeBERTa's Reddit sarcasm AUC rises by 0.022 in en-UK (0.845 to 0.867), which has 248 sarcastic Reddit comments in the pool, and by 0.012 in en-AU (0.833 to 0.845). The embedding probe's official OOF score rises from 0.837 to 0.847. For DeBERTa and the LLM probe, score gains are smaller above 50% than below it; these curves do not estimate the effect of data outside the pool or of models larger than 9 billion parameters.

Averaging training runs. We trained DeBERTa twice with each of seeds 13–17, giving ten training runs. Averaging their predictions yields 0.860 for one run, 0.863 for three and 0.864 for ten (Figure 3b). Adding the two probes raises the score by 0.0069 at three runs and by 0.0058 at ten, more than the 0.0005 gained from three to ten runs. The two five-run ensembles (the original and the retrained runs of seeds 13–17) differ by 0.004, which illustrates how much retraining alone can change the score.

Design choices. We ablated three choices, evaluating predictions averaged over seeds 13 and 14 (baseline 0.865). Removing the `[variety] [source]` prefix lowers the score to 0.857 (suggestive: $P_{\text{boot}} = 0.84$ for the baseline; lower on 4 of 5 folds). Training two single-task models instead of one multi-task model gives 0.865 ($P_{\text{boot}} = 0.63$), and sharing one sarcasm output across varieties gives 0.863 ($P_{\text{boot}} = 0.74$).

6.2 Sarcasm-label rates and cue associations

The two label sets differ in the share of negative comments labelled sarcastic and in how sarcasm labels relate to automatically scored cues. Among Reddit comments labelled sarcastic, 97% (en-AU) and 99% (en-UK) also carry the negative sentiment label, but the en-AU annotator labels 60% of comments with the negative label sarcastic, against 24% in en-UK. The gap holds within each community: among comments that the SiEBERT sentiment classifier (Hartmann et al., 2023) labels negative, each of the three en-AU subreddits has a higher share labelled sarcastic (44–64%) than each of the four en-UK subreddits (12–33%; Figure 1a).

Derision and playful cues. We group our exploratory cues into two kinds, drawing on work that relates sarcasm to ridicule of a target (Lee and Katz, 1998), that models one form of sarcasm as positive sentiment contrasted with a negative situation (Riloff et al., 2013), and that analyses

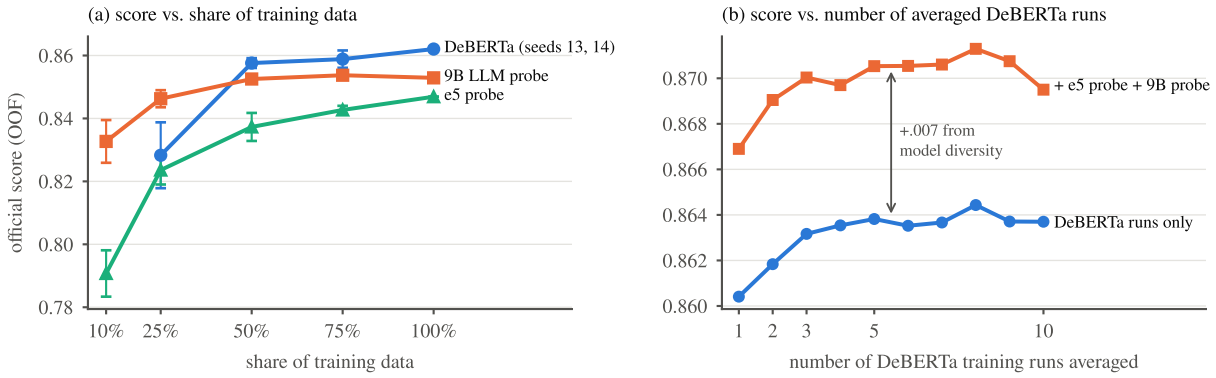


Figure 3: Official OOF score as a function of training-set size and of the number of averaged DeBERTa runs. (a) Each model is trained on nested subsets of every training fold, stratified by segment (DeBERTa: mean of the scores of seeds 13 and 14, evaluated separately; probes: mean of three subset draws below 100%; error bars: standard deviation). (b) Mean score over subsets of ten DeBERTa training runs (two runs at each of seeds 13–17; a run is the five fold models of one training), averaging the runs' predictions (up to 30 subsets per point), alone and with the two probes added. The labelled y-axis ticks run from 0.860 to 0.870: the 0.003 gain from one to three runs is smaller than the 0.004 difference between two five-run ensembles of the same configuration.

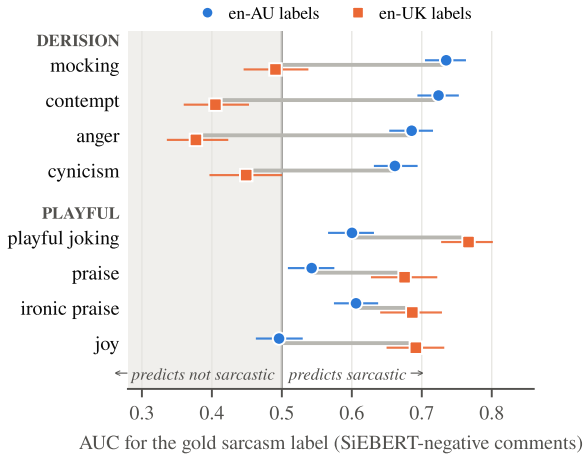


Figure 4: ROC AUC of each derision and playful cue for the annotator's sarcasm label, among Reddit comments that SiEBERT (Hartmann et al., 2023) labels negative (en-AU $n = 1{,}271$, 685 sarcastic; en-UK $n = 912$, 186 sarcastic), with 95% intervals from 1,000 comment-level bootstrap replicates (not adjusted for multiple comparisons). Values above 0.5: higher cue scores among comments labelled sarcastic; below 0.5: lower.

mockery framed as non-serious (Haugh, 2010). We measure each component with automatic cues. *Derision cues* score whether a comment ridicules or expresses hostility towards a target: mocking, contempt, anger and cynicism. *Playful cues* score whether it has a humorous or positive surface: playful joking, praise, ironic praise and joy. The first seven cues are scores from a natural language inference (NLI) classifier (Laurer et al., 2023), which estimates whether the comment entails short hypotheses such as "The author is mocking someone"; joy is the log-probability assigned to joy by an emotion classifier (Hartmann, 2022) (exact hypotheses in Appendix C). Neither model was trained on BESSTIE labels, so the cues are automatic estimates, not human judgements.

Associations with the sarcasm label. We measure each cue's association with the gold sarcasm label as ROC AUC (the probability, counting ties as one half, that a randomly chosen comment labelled sarcastic has a higher cue score than a randomly chosen comment labelled not sarcastic; Figure 4). To avoid selecting comments by the annotator's own sentiment labels, we use the comments that SiEBERT labels negative, and we also report AUCs on all Reddit comments. In en-AU, the derision cues have AUCs of 0.73 (mocking), 0.72 (contempt), 0.69 (anger) and 0.66 (cynicism). In en-UK, contempt and anger have AUCs of 0.41 and 0.38 (inverse associations); mocking has 0.49 (95% interval 0.45–0.54) and cynicism 0.45 (0.40–0.50). The playful cues have AUCs of 0.68–0.77 in en-UK (playful joking 0.77, praise 0.68, ironic praise 0.69, joy 0.69) against 0.50–0.61 in en-AU. On all Reddit comments, the derision AUCs are 0.76–0.80 in en-AU and 0.40–0.51 in en-UK. Restricting again to archive-matched comments that SiEBERT labels negative, the four derision cues have AUCs of 0.62–0.74 within each of the three en-AU subreddits, and contempt and anger have AUCs of 0.33–0.47 in all four en-UK subreddits; en-UK mocking ranges from 0.37 (GreatBritishMemes) to 0.56 (unitedkingdom). The direction

of the contempt and anger associations is therefore consistent across the sampled subreddits, although the two varieties' comments come from different communities. The eight cues were chosen for interpretability from 18 NLI-derived scores (16 hypothesis groups and 2 contrasts between them) and the emotion classifier's outputs; the selection and comparisons are exploratory.

Possible explanations. The guideline counts "irony or mockery" used to express contempt or ridicule. The en-AU associations are compatible with the inclusive reading, in which literal mockery that expresses contempt counts as sarcasm; the en-UK associations are compatible with the narrower reading, in which literal hostility alone does not. These associations do not establish either annotator's decision rule. Possible explanations include differences between the annotators' interpretations of the guideline; differences in how the communities use mockery, which can combine apparent hostility with closeness (Haugh and Bousfield, 2012); and differences between the sampled texts, since each annotator labelled different comments. The cue scores are classifier outputs whose errors may also differ between varieties. Because annotators and texts are not crossed between varieties, these explanations cannot be separated here; a design in which several annotators from each variety label the same comments would allow annotator and item effects to be estimated. Our system's prompt selection is compatible with the same difference: during development, a sarcasm definition that includes literal ridicule raised the AUC of zero-shot Qwen3.5-4B's answer margin on en-AU Reddit comments from 0.65 to 0.78, and a definition that requires non-literal meaning was kept for en-UK (Appendix C); the prompts were chosen by comparing AUCs on the pool, so this is not independent evidence. The difference is consistent with the lower cross-variety sarcasm F-scores that BESSTIE reported. In our own experiment, averaged over three seeds, DeBERTa's Reddit sarcasm AUC on en-AU is 0.840 when trained on en-AU and 0.678 when trained on en-UK (-0.162), and on en-UK it is 0.859 when trained on en-UK and 0.715 when trained on en-AU (-0.144). Sentiment AUC, evaluated on both sources, falls by at most 0.011. Training on both varieties with the variety prefix gives sarcasm AUCs of 0.845 (en-AU) and 0.861 (en-UK), against 0.840 and 0.859 for training on the evaluation variety alone. Our system uses both the variety prefix and variety-specific sarcasm outputs; this comparison does not isolate their separate effects.

6.3 Agreement and sarcasm discrimination in en-AU

BESSTIE's second annotator labelled 50 items per variety (Srirag et al., 2025, §2.3), giving Cohen's κ (agreement corrected for chance; Cohen, 1960) of 0.47 for en-AU sarcasm, against 0.61 (en-AU) and 0.79 (en-UK) for sentiment; BESSTIE reports no confidence interval for these 50-item estimates. Our submitted ensemble's OOF predictions agree with the gold en-AU sarcasm labels at $\kappa = 0.56$ (95% bootstrap interval 0.52–0.59) over all 2,004 en-AU Reddit comments. The two estimates come from different samples and cannot be compared directly, and neither estimates the highest score attainable on this task. Among Reddit comments that the annotator labelled negative, our ensemble's OOF sarcasm AUC is 0.77 in en-AU against 0.91 in en-UK. Our results do not distinguish whether the lower en-AU value reflects the labelling convention, conversational context that the models do not see, or limitations of the models we tested.

7 Interventions That Did Not Meet the Adoption Criterion

After the three-member ensemble was fixed, none of the interventions reported in Appendix A met the adoption criterion of §3. Alternative pre-trained encoders (ELECTRA, ModernBERT and an NLI-tuned DeBERTa) scored 0.009 to 0.016 below DeBERTa-v3-large on the screening fold and changed the ensemble's score by -0.007 to $+0.002$. On the screening fold, an exponential moving average of the weights changed DeBERTa's score by +0.001, layer-wise learning-rate decay with weight averaging by +0.004, continued pre-training on unlabelled in-domain text (domain-adaptive pre-training) by -0.004, and preliminary training on external sarcasm data by -0.011. Qwen3.5-4B fine-tuned with low-rank adaptation (LoRA) scored 0.02 below three-seed DeBERTa on the screening fold and did not change the ensemble's score. Adding recovered thread text (parent comment and submission title and body), features derived from the gold labels of other pool comments in the same thread (cross-fitted), or a label vote of the 30 nearest neighbours by cosine similarity in the embedding probe's representation space changed the score by -0.001, +0.001 and -0.005.

Pseudo-labelling (training on unlabelled texts labelled by our own models) raised the score from 0.881 to 0.886 on the screening fold but lowered it from 0.865 to 0.859 on the confirmation fold ($P_{\text{boot}} = 0.95$ for the baseline), so it failed its pre-specified confirmation.

8 Conclusion

A three-member ensemble selected with cross-validation and paired bootstrap comparisons scored 0.867, ranked second, and was one of six joint winners. In the label analysis, among comments labelled negative by a sentiment classifier, 44–64% are labelled sarcastic in each en-AU subreddit against 12–33% in each en-UK subreddit, and contempt and anger scores are associated with the sarcastic label in en-AU but with the non-sarcastic label in en-UK; BESSTIE reports human agreement of $\kappa = 0.47$ for en-AU sarcasm. Cross-variety performance differences on this benchmark may therefore reflect differences in labelling conventions and texts as well as model generalisation. For our submission, en-AU had the lower macro-F1 in both tasks, and en-AU sarcasm had the lowest test macro-F1 of the four components (0.801). On the pool, among Reddit comments that the annotator labelled negative, our OOF sarcasm AUC was 0.77 in en-AU against 0.91 in en-UK. We suggest that benchmarks comparing varieties specify whether literal mockery that expresses contempt or ridicule counts as sarcasm, have annotators from each variety label a shared set of items, and report agreement per variety.

Limitations

Our models see isolated comments; BESSTIE's guideline asks annotators to consider context, but its annotation description does not specify whether surrounding conversation was shown (Srirag et al., 2025). The exploratory LLM re-annotation (Appendix B) uses one model with a narrower guideline than BESSTIE's and is not evidence of annotation error. With one annotator per variety, each labelling different comments, we cannot separate the effects of individual annotators, communities and texts; our findings describe this dataset's labels, not Australian or British English in general. The human agreement estimate (50 items) and our ensemble's agreement estimate (2,004 comments) come from different samples. Our bootstrap intervals resample comments and do not model dependence between comments from the same thread. The cue analyses are exploratory, use classifier scores as indirect measures of derision and playful language without manual validation in each variety, and are not corrected for multiple comparisons. We did not test models larger than 9 billion parameters, so our curves do not rule out gains from larger models. The bootstrap standard deviation of the official score at development-file size is about 0.013, larger than 13 of the 17 changes in Table 3, so we base comparisons on OOF predictions and paired resampling. The DeBERTa epoch count was chosen on the fold later used for screening, which may bias those screens upwards. The system always labels en-UK reviews not sarcastic.

Ethics and Data

Our code is released at `https://github.com/Sekinal/alta2026-stargazers`, and a release with predictions, cached features and model checkpoints at `https://huggingface.co/Thermostatic/alta2026-thermostatic`. Neither release contains competition texts; Table 1 contains paraphrases. For the split audit and the thread-context experiments we matched Reddit comments to a public archive and used subreddit names, thread identifiers, timestamps and the text of parent comments and submissions, without usernames; we release none of this recovered material. Competition texts were processed by Anthropic's models through Claude Code during development. The LLM re-annotation (Claude Opus 5.5) received the guideline, the text, its variety and an item identifier, without gold labels or model predictions. The GPT-6 Astra reviewers received the paper and aggregate results, not competition texts. External data used in exploration came from public sources under the terms listed in Appendix D; none is redistributed.

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All code, experiments and analyses, and the first draft of every section of this paper, were produced by Claude Code (Anthropic, 2026), running Claude Opus 5 and Claude Opus 5.5, under the author's direction; separate Claude Opus 5.5 instances served as the LLM annotators (Appendix B) and as internal reviewers, and OpenAI's GPT-6 Astra, run through the Codex command-line interface, reviewed the paper in several rounds. The design drew on the author's experience in Kaggle compe-

titions, including a pseudo-labelling procedure and a classification-head configuration adapted from the author's earlier Kaggle competition system (AstroCLIMB), and on a ChatGPT-generated list of LoRA variants with links to their papers that the author supplied; the agents contributed the ensemble, output-head, prefix and probe designs from their prior knowledge. The sarcasm and annotation literature was read after the system was fixed and did not change it. A full account of the development process, including mistakes found and corrected, is released with our code. The author reviewed and edited the paper and takes full responsibility for its content; artificial intelligence (AI) systems are not authors of this work. We thank Lium for a generous grant of GPU (graphics processing unit) credits that made this and other work possible, and the organisers, Diego Mollá-Aliod, Aditya Joshi and Dipankar Srirag, for running the shared task and for their careful significance analysis.

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A Results of Additional Interventions

Table 3 lists the interventions we evaluated after the three-member ensemble was fixed, under the protocol of §3. Δ is the intervention's official score minus its baseline's. One-fold DeBERTa changes compare predictions averaged over the same three seeds; LoRA changes compare mean scores of individual runs (two seeds, except one seed for EVA + DoRA + rsLoRA + Riemannian); five-fold changes compare OOF scores on the full pool.

B Exploratory LLM Re-annotation

We asked whether an LLM annotator without access to the gold labels would reproduce them. Two separate runs of Claude Opus 5.5 labelled the same 399 Reddit comments: 150 sampled uniformly per variety, plus 50 en-AU and 49 en-UK *disputed* comments, whose OOF ensemble sarcasm probability differed from the gold sarcasm label by more than 0.8. Each run received the comments in shuffled batches of 100, with the text, its variety and an item identifier but no labels or predictions, and this guideline: "You are annotating short Reddit comments from Australian (en-AU) or British (en-UK) English speakers. For each item give two labels, judging the text as written (you do not see the thread it came from): sarcasm: 1 if the author says something they do not literally mean in order to mock, criticise, or be ironic/teasing (sarcasm, verbal irony, mock praise, rhetorical ridicule); 0 otherwise. Plain complaints, jokes that are not ironic, and sincere statements are 0. sentiment: 1 if the overall attitude/emotion expressed is positive, 0 if it is negative (or neutral-leaning negative)." This guideline is narrower than BESSTIE's: it requires non-literal meaning and counts plain complaints as not sarcastic. The sample, guideline and criteria were written before annotation and committed to version control together with the results.

Results. On the 300 uniformly sampled comments, the two runs agree with each other at $\kappa = 0.90$ (en-AU) and 0.97 (en-UK) for sarcasm and 0.91 and 0.98 for sentiment. This measures agreement between repeated runs of one model, not agreement with independent human judgements. LLM labels can also differ systematically from human judgements on subjective tasks (Pavlovic and Poesio, 2024). In the uniform samples, 43% of en-AU comments are labelled sarcastic in the gold data against 16% by the LLM runs (mean), and 29% against 27% in en-UK (Table 4). Our ensemble agrees with the gold en-AU sarcasm labels at $\kappa = 0.56$ and with the two runs at 0.30 and 0.34.

Pre-specified criteria. We wrote down three criteria for the hypothesis that label disagreement limits further gains: evaluated on sarcasm, separately

Intervention (protocol)	Δ
DeBERTa training / encoder (one fold, same 3 seeds)	
weight averaging (EMA)	+.001
layer-wise learning-rate decay + EMA	+.004
NLI-tuned DeBERTa-v3-large	−.016
ELECTRA-large	−.009
ModernBERT-large	−.012
any of these added to the ensemble	−.007 to +.002
continued in-domain pre-training	−.004
sarcasm pre-training (external data)	−.011
LLMs (Qwen3.5-4B; Qwen Team, 2026; one fold)	
LoRA + rsLoRA, alone / in ensemble	−.02 / .000
EVA + rsLoRA	−.04
EVA + DoRA + rsLoRA + Riemannian (1 seed)	−.10
Data and context (five folds unless noted)	
pseudo-labels (pre-spec., fold 4)	−.006
recovered thread text (stacked)	−.001
same-thread label features	+.001
30-nearest-neighbour label vote	−.005
LLM label as feature, en-UK (pre-spec.)	.000

Table 3: Change in official score against the matching baseline (standalone DeBERTa rows and LoRA alone: three-seed DeBERTa on the same fold; "added to the ensemble" and "in ensemble": the unchanged ensemble on the same fold; LoRA variants: LoRA + rsLoRA; pseudo-labels: the unchanged ensemble on fold 4; other five-fold rows: the full ensemble, or for thread context the same second-stage classifier without the thread features). Pre-spec.: evaluated against a criterion written before the run. Methods: exponential moving average of the weights (EMA); layer-wise learning-rate decay (Howard and Ruder, 2018); NLI-tuned DeBERTa (Laurer et al., 2024); ELECTRA (Clark et al., 2020); ModernBERT (Warner et al., 2025); continued in-domain pre-training (Gururangan et al., 2020); sarcasm pre-training data (Appendix D); low-rank adaptation (LoRA; Hu et al., 2022) with rank-stabilised scaling (rsLoRA; Kalajdzievski, 2023), explained variance adaptation (EVA; Paischer et al., 2025), used to initialise the adapters, weight-decomposed LoRA (DoRA; Liu et al., 2024) and Riemannian preconditioning (Zhang and Pilanci, 2024), implemented with the parameter-efficient fine-tuning (PEFT) library (Mangrulkar et al., 2022) (the EVA initialisation was capped at 32 batches, with 115 and 127 of 248 layers not converged for seeds 0 and 1); pseudo-labelling (Lee, 2013): a student model first trained on 400k unlabelled in-domain texts labelled by our fold models, then fine-tuned on the gold labels. For the alternative encoders, sarcasm pre-training, the LoRA variants and pseudo-labelling (0.95), P_{boot} for the baseline is at least 0.8; no intervention reaches $P_{\text{boot}} \geq 0.8$ over its baseline.

per variety: (i) the two runs agree with each other more than with gold on the uniform sample (met); (ii) our ensemble agrees with the runs at least as much as with gold on the uniform sample (not met in either variety); (iii) on the disputed comments where the runs agree, they agree with our ensemble more than 60% of the time (met: 31 of 45 en-AU, 31 of 48 en-UK). Criterion (iii) does not distinguish agreement with the ensemble from a preference for the non-sarcastic label: on en-AU comments the runs choose the non-sarcastic label in 28 of 32 cases where the ensemble gives it and in 10 of 13 cases where the gold label gives it. Among the 48 en-UK disputed comments on which the runs agree, they agree with the ensemble on 26 of 42 where it predicts not sarcastic and on 5 of 6 where it predicts sarcastic. The pre-specified evidence therefore does not establish that label disagreement limits further gains. In a separate pre-specified experiment, one run labelled all 1,143 en-UK Reddit comments in the pool; a cross-fitted logistic regression on the ensemble's log-odds and this label raised en-UK Reddit sarcasm AUC from 0.906 to 0.920 and en-UK sarcasm macro-F1 from 0.841 to 0.843, but left the official score unchanged, because en-AU remained the binding variety.

	G–O	G–R	O–R	Positive rate G / O / R
Sarc AU	**.56**	.21–.28	.30–.34	.43 / .43 / .16
Sarc UK	.59	.62	.45	.29 / .23 / .27
Sent AU	.74	.60–.65	.60–.65	.29 / .33 / .26
Sent UK	.85	.81–.83	.67–.69	.17 / .15 / .22

Table 4: Agreement (Cohen's κ, Cohen, 1960) on 150 random Reddit comments per variety between the gold labels (G), our ensemble (O) and the two readers (R; ranges over readers), and the share each labels positive (sarcastic, or positive sentiment; R is the mean of the two runs). 95% bootstrap intervals for G–O sarcasm: 0.43–0.70 (en-AU) and 0.44–0.72 (en-UK).

Limitations of the re-annotation. Both runs used one LLM, from the model family used to develop our system, with a stricter definition than BESSTIE's; they are not judgements by speakers of either variety. The lower en-AU rate (16% against 43%) may reflect the stricter definition, a model that does not recognise jocular mockery and abuse, which have been described in Australian and British interaction (Haugh and Bousfield, 2012), or the annotator's convention; we cannot separate these. The analyses in the main text do not use these labels.

C Reproducibility Details

Folds and thresholds. Five folds over the pool (fold-assignment seed 2026), stratified by segment and both labels, with identical texts kept in one fold. Thresholds: one per task and segment, chosen by coordinate ascent (one threshold at a time) to maximise the macro-F1 of the corresponding task and variety; en-UK reviews are always labelled not sarcastic. Reported OOF scores fit thresholds on four folds and apply them to the fifth; test thresholds are fitted on all OOF predictions.

Ensemble. On the test set, the DeBERTa member's probability is the mean over its 25 models (5 seeds $\times$ 5 folds); for OOF predictions it is the mean over the five seeds trained without the text's fold. Each probe's test probability is the mean over its five fold models. With $\ell(p) = \log \frac{p}{1-p}$ and $\sigma(z) = 1/(1+e^{-z})$, the ensemble probability is $\sigma\big(\frac{1}{3}[\ell(p_{\mathrm{DeB}}) + \ell(p_{\mathrm{e5}}) + \ell(p_{\mathrm{LLM}})]\big)$.

DeBERTa. The first token's final hidden state feeds one sentiment output and two sarcasm outputs, one per variety (the text's variety selects the sarcasm output); loss = binary cross-entropy for sentiment + binary cross-entropy for sarcasm, equal weights; AdamW (Loshchilov and Hutter, 2019), learning rate 10^{-5}, batch 16, linear warmup over the first 10% of steps, bfloat16 arithmetic, 4 epochs, output-layer learning rate 10^{-4}, maximum length 256 tokens including the two special tokens (longer texts keep their first 127 and last 127 tokens); seeds 13–17.

Embedding probe. e5-large-v2 embeddings with the prefix "query: " are concatenated with multilingual-e5-large-instruct embeddings computed with the instruction "Decide whether the given online review or comment is sarcastic or ironic"; features are standardised; sentiment uses one logistic regression with segment indicators, sarcasm one logistic regression per segment (en-UK reviews receive the pool prevalence), with inverse regularisation strength C chosen by three-fold inner cross-validation.

LLM probe. Qwen3.5-9B with 4-bit NormalFloat (NF4) weights (double quantisation, bfloat16 arithmetic), using its chat template with the reasoning mode disabled. Features: the layer-24 hidden state at the last prompt token, the answer margin (log-sum-exp of the logits of "Yes", "yes", "YES" minus that of "No", "no", "NO"; Positive/Negative variants for sentiment), and indicators for en-UK, Reddit and their interaction; standardised; logistic regression (inverse regularisation strength $C = 0.002$), with one sarcasm probe per variety (the en-UK sarcasm probe excludes reviews). The prompt names the text type (e.g. "a Reddit comment from an Australian community (Australian English)"), gives a definition and asks "Is the text sarcastic? Answer Yes or No." The en-AU definition reads: "Sarcasm here is understood broadly: the writer uses irony, mockery, ridicule, cynical or derisive jabs, snark, or says something they do not literally mean, usually to criticise or show contempt (for example cynical remarks about politicians, parties or government). A sincere, neutral, informative or friendly text is NOT sarcastic." The en-UK definition reads: "Sarcasm means the writer says something they do not literally mean, usually the opposite, in order to mock, criticise or show contempt. [...] A text that is merely angry, negative, rude or critical but says exactly what it means is NOT sarcastic. British sarcasm is often dry and understated [...]"; the elided parts list example forms and phrases. The sentiment prompt asks whether the overall sentiment is positive or negative and states that sarcastic praise counts as negative. Full prompts are in `experiments/llm/llm_common.py` in the release.

Cue scores. Each NLI cue is the mean entailment log-odds of `MoritzLaurer/deberta-v3-large-zeroshot-v2.0` over two or three hypotheses, with the comment as premise (256 tokens, head and tail): *mocking*: "The author is mocking someone.", "The author is ridiculing something.", "This text makes fun of someone."; *contempt*: "The author is contemptuous.", "The author looks down on someone."; *anger*: "The author is angry.", "The author is frustrated.", "The author is annoyed."; *cynicism*: "The author is cynical.", "The author expects things to go badly."; *playful joking*: "The author is joking.", "This text is meant to be funny.", "The author is being playful."; *praise*: "The author is praising something.", "The author is happy with something.", "This is a compliment."; *ironic praise*: "This is ironic praise.", "The author pretends to be positive but is actually negative.", "The compliment in this text is not meant seriously." *Joy* is the emotion classifier's log-probability of joy. The other nine NLI hypothesis groups (opposite meaning, sincerity, complaint, politics,

exaggeration, rhetorical question, positive, negative, advice) and two contrasts (praise minus complaint, opposite meaning minus sincerity) were not used as cues; we chose the eight shown as interpretable markers of derision and play, and the selection is exploratory. AUC intervals use 1,000 comment-level bootstrap replicates.

D Resources

Final system. `microsoft/deberta-v3-large` (He et al., 2023); `intfloat/e5-large-v2` (Wang et al., 2022); `intfloat/multilingual-e5-large-instruct` (Wang et al., 2024); `Qwen/Qwen3.5-9B` (Qwen Team, 2026) with bitsandbytes 4-bit (NF4) quantisation (Dettmers et al., 2023); PyTorch (Paszke et al., 2019), Transformers (Wolf et al., 2020), sentence-transformers (Reimers and Gurevych, 2019) and scikit-learn (Pedregosa et al., 2011); data processing with Polars (Vink et al., 2024) and figures with Matplotlib (Hunter, 2007). Trained only on the shared-task training and development data. Checkpoints screened in Table 3: `google/electra-large-discriminator`, `answerdotai/ModernBERT-large`, `MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli` and `Qwen/Qwen3.5-4B`.

Analyses. `MoritzLaurer/deberta-v3-large-zeroshot-v2.0` (Laurer et al., 2023), `siebert/sentiment-roberta-large-english` (Hartmann et al., 2023) and `j-hartmann/emotion-english-distilroberta-base` (Hartmann, 2022) for the cues of Figure 4; SiEBERT supplies the sentiment filter (the NLI model produces seven cues and the emotion classifier the joy cue); Reddit thread metadata matched through Arctic Shift (Heitmann, 2024) for the split audit.

External data in the reported negative results. The sarcasm pre-training of Table 3 used Reddit comments that their authors marked as sarcastic with the tag `/s` (tag removed) from 24 AU and 26 UK subreddits (listed in the release), collected through Arctic Shift, ConvoKit's per-subreddit corpora (Chang et al., 2020) and the UK subreddit dumps of alujjdnd (2023), with untagged comments from the same subreddits assigned the non-sarcastic label; it added EPIC (Frenda et al., 2023), the English intended-sarcasm data of iSarcasmEval (Abu Farha et al., 2022), the English irony tweets of SemEval-2018 Task 3 (Van Hee et al., 2018), the Reddit and Twitter sarcasm data of the FigLang 2020 shared task (Ghosh et al., 2020) and the Reddit irony corpus of Wallace et al. (2014). Domain-adaptive pre-training and pseudo-labelling used unlabelled in-domain text: AU and UK Reddit via Arctic Shift (Reddit's terms), Google Local reviews (He et al., 2017; Pasricha and McAuley, 2018), Booking.com reviews (`huggingface.co/datasets/Booking-com/accommodation-reviews`, Creative Commons Attribution-ShareAlike 4.0), reviews of National Health Service general practices in England (`huggingface.co/datasets/janduplessis886/england-nhs-gp-reviews`, Open Government Licence) and Inside Airbnb reviews (`insideairbnb.com`, Creative Commons Attribution 4.0). All external data were deduplicated against the shared-task training and development pool, and none is used by the final system.

Other exploration. We also screened other encoders, embedding models, LLMs, zero-shot feature models and classical models (logistic regression and support vector machines on term-frequency–inverse-document-frequency features, and gradient-boosted trees), and collected further sarcasm and sentiment corpora. None contributed to the final system; the only one used in a reported result is the BGE-large embedding model, in the calibration run of §3; `results/paper/PROCESS.md` in the release lists them all with their sources.